# MATHEMATICS

**TARGET : JEE-ADVANCED-2024**

# CAPS-12$^{+}$

**SCQ (Single Correct Type) :**

**1.** The area defined by $|y| \le e^{-|x|} - \frac{1}{2}$ in cartesian co-ordinate system, is :

(A) (4 – 2 1n 2) (B) (4 – In 2) (C) (2 – In 2) (D) (2 – 2 1n 2)

**2.** For each positive integer n > 1; $A_n$ represents the area of the region restricted to the following two inequalities : $\frac{x^2}{n^2} + y^2 \le 1$ and $x^2 + \frac{y^2}{n^2} \le 1$. Find $\lim\limits_{n \to \infty} A_n$.

(A) 4 (B) 1 (C) 2 (D) 3

**3.** Point A lies on curve $y = e^{-x^2}$ and has the coordinate (x, $e^{-x^2}$) where x > 0. Point B has coordinates (x, 0). If 'O' is the origin, then the maximum area of $\Delta$AOB is :

(A) $\frac{1}{\sqrt{8e}}$ (B) $\frac{1}{\sqrt{4e}}$ (C) $\frac{1}{\sqrt{2e}}$ (D) $\frac{1}{\sqrt{e}}$

**4.** Area bounded by $x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$ is equal to :

(A) $\frac{4\pi}{3} + \sqrt{2}$ (B) $\frac{4\pi}{3} - \sqrt{2}$ (C) $\frac{4\pi}{3} + 2\sqrt{3}$ (D) None of these

**5.** A circle centered at origin and having radius $\pi$ units is divided by the curve y = sin x in two parts. Then area of the upper part equals to :

(A) $\frac{\pi^2}{2}$ (B) $\frac{\pi^3}{4}$ (C) $\frac{\pi^3}{2}$ (D) $\frac{\pi^3}{8}$



**MCQ (One or more than one correct) :**

**6.** Let $I_n = \int_0^n 2^x(1 + \{g(x)\} + \{\sqrt{g(x)}\})dx, n \in N$, where g(x) = $\sin^2(\pi x)$ and {x} represents fractional part of x, then

(A) $I_1 > \log_2 e$ (B) $I_1 < \log_2 e$ (C) $I_4 = 15I_1$ (D) $I_4 = 16I_1$

**7.** Let b > a > 0, $I_1 = \int_a^{a+b/2} \left(\sqrt{\frac{b-x}{x-a}} - \sqrt{\frac{x-a}{b-x}}\right) dx$ and $I_2 = \int_{\frac{a+b}{2}}^{b} \left(\sqrt{\frac{x-a}{b-x}} - \sqrt{\frac{b-x}{x-a}}\right) dx$, then

(A) $I_1 = I_2$

(B) $I_1 > I_2$

(C) $I_1 < I_2$

(D) $I_1 + I_2 = 10$ if b – a = 5

**8.** The value of the definite integral $\int_{-\infty}^{a} \frac{(\sin^{-1} e^x + \sec^{-1} e^{-x})dx}{(\cot^{-1} e^a + \tan^{-1} e^x)(e^x + e^{-x})}$ $(a \in R)$ is

(A) Independent of a

(B) dependent on a

(C) $\frac{\pi}{2}\ln 2$

(D) $-\frac{\pi}{2}\ln\left(\frac{2}{\pi}\tan^{-1} e^{-a}\right)$

**9.** If f(x) is a differentiable function such that f(x + y) = f(x) f(y) $\forall$ x, y $\in$ R, f(0) $\neq$ 0 and $g(x) = \frac{f(x)}{1+(f(x))^2}$,

then

(A) $\int_{-2014}^{2015} g(x)dx = \int_{0}^{2015} g(x)dx$

(B) $\int_{-2014}^{2015} g(x)dx - \int_{0}^{2014} g(x)dx = \int_{0}^{2015} g(x)dx$

(C) $\int_{-2014}^{2015} g(x)dx = 0$

(D) $\int_{-2014}^{2014} 2g(-x) - g(x)dx = 2\int_{0}^{2014} g(x)dx$

**10.** Consider, $f(x) = \begin{cases} x^{\frac{1}{3}}; & 0 \le x < 1 \\ (x-2)^2; & 1 \le x < 2 \end{cases}$ and f(–x) = f(x) $\forall x \in R$ and f(x + 2) = f(x) $\forall$ x $\geq 0$, the identify the

correct statement(s)

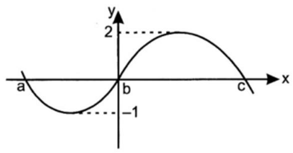
(A) The number of points of non-derivability of f(x) in [0, 10] is(are) 11.

(B) The number of points of non-derivability of f(x) in [0, 10] is(are) 9.

(C) Area of the region enclosed by the f(x) and the x-axis from x = –1 to x = 7 is $\frac{19}{4}$ sq. units.

(D) Area of the region enclosed by the f(x) and the x-axis from x = –1 to x = 7 is $\frac{49}{12}$ sq. units.

**11.** Let f(x) be a polynomial function of degree 3 where a < b < c and f(a) = f(b) = f(c). If the graph of f(x) is as shown, which of the following statements are **INCORRECT** ? (Where c > |a|)




(A) $\int_a^c f(x)dx = \int_b^c f(x)dx + \int_a^b f(x)dx$ (B) $\int_a^b f(x)dx < 0$

(C) $\int_a^b f(x)dx < \int_c^b f(x)dx$ (D) $\frac{1}{b-a}\int_a^b f(x)dx > \frac{1}{c-b}\int_b^c f(x)dx$

**Numerical Based Questions :**

**12.** Let the function f : [–4, 4] → [–1, 1] be defined implicitly by the equation $x + 5y - y^5 = 0$. If the area of triangle formed by tangent and normal to f(x) at x = 0 and the line y = 5 is A, find $\frac{A}{13}$.

**13.** Area of the region bounded by $[x]^2 = [y]^2$, if $x \in [1, 5]$, where [ ] denotes the greatest integer function, is :

**Paragraph Type Question :**

**Paragraph for Question Nos. 14 to 16**

Suppose f(x) and g(x) are differentiable functions such that x g(f(x)). f'(g(x)).g'(x)= f(g(x)).g'(f(x)) f'(x) $\forall$ x $\in$ R and f(x) & g(x) are positive for all x $\in$ R. Also $\int_0^x f(g(t))dt = \frac{1}{2}(1-e^{-2x})\forall x \in R$, g(f(0)) = 1 and h(x) = $\frac{g(f(x))}{f(g(x))}\forall x \in R$.

**14.** The graph of y = h(x) is symmetric with respect to the line:

(A) x = –1 (B) x = 0 (C) x = 1 (D) x = 2

**15.** The value of f(g(0)) + g(f(0)) is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

**16.** The largest possible value of h(x) $\forall$ x $\in$ R is

(A) 1 (B) e1/3 (C) e (D) e2

## Paragraph for Question Nos. 17 to 19

For j = 0, 1, 2, ... n let $S_j$ be the area of region bounded by the x-axis and the curve $ye^x = \sin x$ for $j\pi \le x \le (j + 1)\pi$

**17.** The value of $S_0$ is :

(A) $\frac{1}{2}(1+e^{\pi})$ (B) $\frac{1}{2}(1+e^{-\pi})$ (C) $\frac{1}{2}(1-e^{-\pi})$ (D) $\frac{1}{2}(e^{\pi}-1)$

**18.** The ratio $\frac{S_{2009}}{S_{2010}}$ equals :

(A) $e^{-\pi}$ (B) $e^{\pi}$ (C) $\frac{1}{2}e^{\pi}$ (D) $2e^{\pi}$

**19.** The value of $\sum_{j=0}^{\infty} S_j$ equals to :

(A) $\frac{e^{\pi}(1+e^{\pi})}{2(e^{\pi}-1)}$ (B) $\frac{1+e^{\pi}}{2(e^{\pi}-1)}$ (C) $\frac{1+e^{\pi}}{e^{\pi}-1}$ (D) $\frac{e^{\pi}(1+e^{\pi})}{(e^{\pi}-1)}$

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### ANSWERKEY OF CAPS – 12+

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1.** | (D) | **2.** | (A) | **3.** | (A) | **4.** | (C) | **5.** | (C) | **6.** | (AC) | **7.** | (AD) |
| **8.** | (BD) | **9.** | (BD) | **10.** | (AC) | **11.** | (CD) | **12.** | 5 | **13.** | 8 | **14.** | (C) |
| **15.** | (B) | **16.** | (C) | **17.** | (B) | **18.** | (B) | **19.** | (B) | | | | |